Basic Configuration for First-Time Use

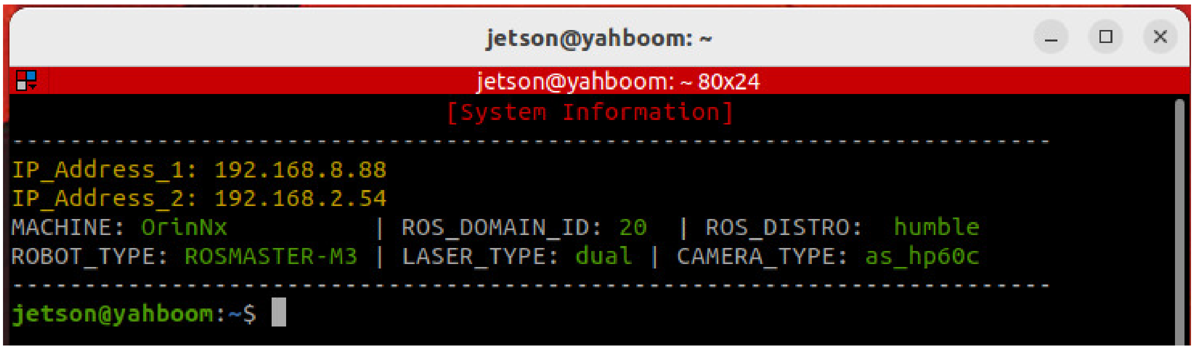
1. Course Content

- Steps required for users to configure the machine for the first time.

2. Configure LiDAR Type

The ROSMASTER-M3 distinguishes between single and dual LiDAR versions. You need to modify the system environment variables according to your car's LiDAR type.

- Open a terminal, and it will print basic system information.



Meaning of terminal information:

- `MACHINE`: Car's mainboard type
- `ROBOT_TYPE`: Product robot model
- `ROS_DOMAIN_ID`: ROS domain ID
- `ROS_DISTRO`: ROS version
- `LASER_TYPE`: LiDAR type, distinguishing between single and dual LiDAR versions
- `CAMERA_TYPE`: Distinguishes between Orbbec Dabai_DCW2 and ANSEE Nuwa-HP60C versions

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first:

```
bringup_m3
```

Open a terminal inside the container:

```
exec_m3
```

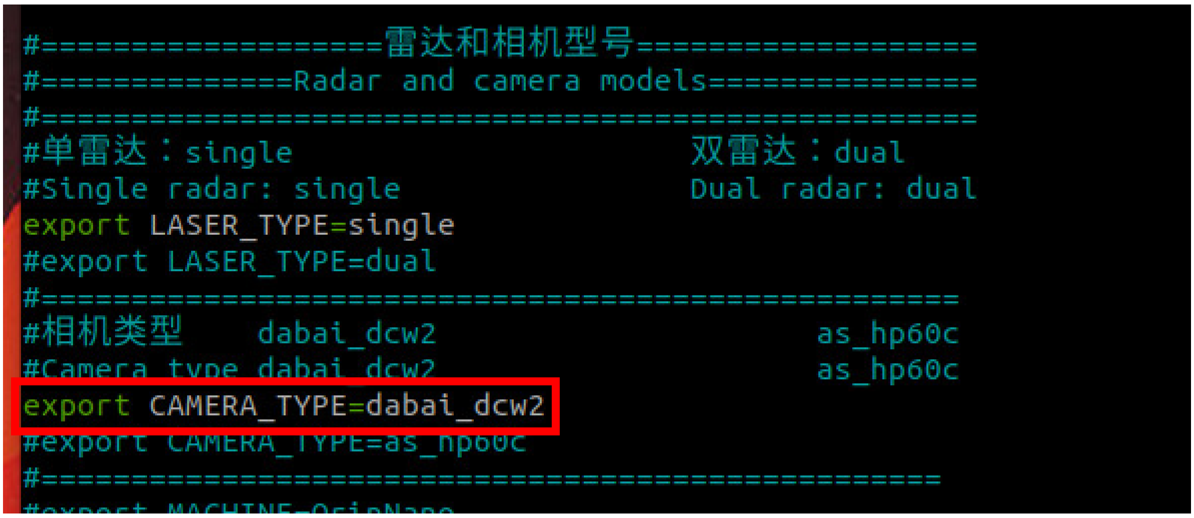
- Open the .bashrc file to modify `LASER_TYPE`:

```
nano ~/.bashrc
```

- Uncomment the corresponding line based on your car's single or dual LiDAR type. The following image shows an example for a single LiDAR version.
- For a single LiDAR version, uncomment `export LASER_TYPE=single` and comment out `#export LASER_TYPE=dual`.
- For a dual LiDAR version, uncomment `export LASER_TYPE=dual` and comment out `#export LASER_TYPE=single`.

3. Configure Camera Type

- Similarly, modify `CAMERA_TYPE` to your camera type. The following image shows an example for the Dabai_DCW2 camera.
- Orbbec Dabai_DCW2: `dabai_dcw2`
- ANSEE Nuwa-HP60C: `as_hp60c`



- After modifying, press `Ctrl+X`, then enter `y` to save and exit.
- Refresh the environment variables:

```
source .bashrc
```

Check that `LASER_TYPE` and `CAMERA_TYPE` in the terminal information match your LiDAR and camera types.